

# First-Cut Piping Load Estimation

*Full technical note for mass, gravity, tributary support screening, analytical beam sag and sustained-stress screening without LFEA*

| Repository basis<br>Advanced_Analysis                                                                                                                                                                                                                                                                                                                                                                  | Baseline<br>main @ 6ca233bf... | Method class<br>Preliminary screening | Status<br>Implementation specification |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------|----------------------------------------|
| <b>Authority statement:</b> this method is a preliminary engineering screening tool. It may estimate mass, gravity load paths, simple-span or qualified one-dimensional continuous-beam gravity effects, sag and a sustained-stress screening indicator. It shall not claim solved thermal, guide, line-stop, anchor, nozzle, spring, gap, lift-off, friction, occasional or expansion-stress results. |                                |                                       |                                        |

## 1. Purpose

This technical note defines a complete, traceable and fail-closed first-cut piping load-estimation method for use before a full LFEA analysis is available. The method is intended to support early civil-load coordination, support-layout review, mass reconciliation, gravity screening, span/sag checks, data-completeness review and identification of systems that require escalation.

The method is not a substitute for a three-dimensional piping flexibility analysis. Its outputs shall always retain the method name, applicability status, assumptions, source data, load case and escalation status.

## 2. Governing scope

| Capability                                        | First-cut authority | Boundary                                                                                                       |
|---------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------|
| Bulk engineering-data enrichment                  | YES                 | Resolve or stage missing section, material, fluid, insulation, component-mass and support-capability evidence. |
| Pipe, fluid and insulation mass                   | YES                 | Closed-form geometry and user-authorized density data.                                                         |
| Component point mass and COG                      | YES                 | Explicit catalogue/vendor/project master or accepted user override.                                            |
| EMPTY/OPE/HYD gravity primitives                  | YES                 | Case-specific readiness and no hidden fluid/insulation assumptions.                                            |
| Simple-chain tributary support screening          | YES                 | Existing W10.5 simple-span method and exact equilibrium.                                                       |
| One-dimensional continuous-beam gravity screening | CONDITIONAL         | Only qualified planar/ordered beam paths with bilateral vertical supports and no nonlinear effects.            |
| Sag screening                                     | YES                 | Method-specific deflection; project criterion supplied by user/project profile.                                |
| Sustained stress screening indicator              | CONDITIONAL         | Gravity/pressure indicator only; no compliance claim.                                                          |
| Thermal support loads                             | NO                  | Requires solved structural stiffness and thermal strain.                                                       |
| Guide and line-stop loads                         | NO                  | Requires solved lateral/longitudinal restraint response.                                                       |
| Anchor loads and moments                          | NO                  | Requires six-DOF reaction recovery.                                                                            |
| Nozzle loads                                      | NO                  | Requires equipment-interface reaction recovery and local basis.                                                |
| Lift-off, gaps and friction                       | NO                  | Nonlinear contact problem.                                                                                     |
| ASME B31.3 compliance                             | NO                  | Only a screening indicator may be shown; qualified code evaluation belongs to LFEA B-4.0.                      |

### 2.1 Required result labels

- FIRST-CUT MASS ESTIMATE
- SCREENED GRAVITY SHARE
- ONE-DIMENSIONAL GRAVITY BEAM SCREENING
- SAG SCREENING
- SUSTAINED STRESS SCREENING INDICATOR
- NOT EVALUATED - RUN LFEA

### 2.2 Prohibited result labels

- Reaction, unless the value comes from LFEA B-3.4 or another qualified structural solution.
- Thermal load, guide load, line-stop load, anchor load, nozzle load or lift-off load.
- B31.3 compliant, code passed, code failed or safe.
- Exact continuous-beam coefficient, exact lift-off multiplier or exact thermal redistribution.

### 3. Architecture and authority separation

Imported source / Sequential Sketcher source

- > enrichment candidate set
- > accepted override and assumption set
- > enriched engineering projection
- > W10.4 mass and gravity-load primitives
- > topology qualification

Branch A:

- W10.5 simple-chain tributary screening
- > screened vertical support shares

Branch B:

- first-cut 1D gravity beam screening
- > vertical beam reactions, span moments and sag
- > optional sustained-stress screening indicator

Presentation:

- SVG / WebGL callouts and calculation-basis panel
- > read-only consumption of sealed results

Escalation:

- unsupported topology or load physics
- > LFEA mechanical model, solve, recovery and code evaluation

#### 3.1 Existing repository owners

| Owner                  | Authority                                   | Requirement                                                                                      |
|------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------|
| W10.3                  | Support attachment and restraint capability | Determines whether the vertical direction is RESTRAINED, FREE, GAP, SPRING, UNKNOWN or CONFLICT. |
| W10.4                  | Mass and model-load primitives              | Owns pipe, contents, insulation and component mass and EMPTY/OPE/HYD gravity primitives.         |
| W10.5                  | Simple-chain tributary screening            | Owns simple-span screened vertical forces and equilibrium within its declared scope.             |
| First-cut beam package | Optional new screening authority            | Owns only qualified one-dimensional gravity beam moments, reactions and deflection.              |
| LFEA B-3.3/B-3.4       | Structural solution and recovery            | Owns actual displacement, reaction, element actions and component resultants.                    |
| LFEA B-4.0             | ASME B31.3 evaluation                       | Owns qualified code evaluation with caller-supplied edition/factor datasets.                     |

### 4. Input data and evidence hierarchy

**No missing engineering value may become a calculation input merely because the UI contains a default field. Every numerical input must be explicit source evidence, an accepted user override, a user-authorized dataset value, or a traceable user-approved approximation.**

#### 4.1 Data hierarchy

| Priority | Source                                 | Rule                                                                                     |
|----------|----------------------------------------|------------------------------------------------------------------------------------------|
| 1        | Explicit accepted source evidence      | Imported geometry, material, mass, COG, fluid, insulation or support evidence.           |
| 2        | Accepted user override                 | Staged and explicitly accepted value with source, reason, scope and affected identities. |
| 3        | User-authorized project/master dataset | Versioned line list, material database, catalogue or component-weight master.            |
| 4        | User-approved approximation            | Separate status, source, approver, limitations and semantic identity.                    |
| 5        | BLOCKED / UNRESOLVED                   | Required when none of the above is available.                                            |

#### 4.2 Required input groups

| Group      | Minimum evidence                                                                                    |
|------------|-----------------------------------------------------------------------------------------------------|
| Geometry   | Node coordinates, ordered component path, component lengths, pipe OD and wall or section reference. |
| Material   | Density for mass; E and section inertia for optional beam sag screening.                            |
| Contents   | Case-specific fluid density; explicit zero is allowed and differs from missing.                     |
| Insulation | Thickness and density, or explicit mass per length; explicit zero is allowed.                       |
| Components | Mass, COG or qualified centre, source and revision.                                                 |

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| Group             | Minimum evidence                                                                       |
|-------------------|----------------------------------------------------------------------------------------|
| Supports          | Attachment identity, station and vertical restraint capability.                        |
| Gravity           | Magnitude, direction and project coordinate profile.                                   |
| Pressure          | Only for optional sustained stress screening; pressure basis and source are mandatory. |
| Screening profile | Method, tolerances, sag criterion, permitted approximations and escalation rules.      |

### 4.3 Hidden defaults prohibited

- Default pipe size, wall thickness, schedule, material density or span length.
- Default water density or default insulation mass/density.
- Default operating or design temperature.
- Rating-dependent valve, flange, tee or reducer mass formula presented as ASME-derived.
- Default support type when the directional capability is unresolved.
- Default mill tolerance or corrosion deduction.
- Default allowable stress, SIF, stress index or code factor.

## 5. Topology and support qualification

### 5.1 W10.5 simple-chain screening

- The path is connected, acyclic and has exactly two terminal ports.
- No branch, tee, cross or loop exists in the screened connected component.
- Components can be ordered into one continuous path.
- Every eligible support has one qualified path attachment and a deterministic station.
- The vertical direction is explicitly or validly classified as RESTRAINED.
- GAP, SPRING, UNKNOWN and CONFLICT vertical states block the support or the affected path.
- Loads outside the first-to-last support envelope remain blocked as unsupported overhangs.

### 5.2 Optional one-dimensional continuous-beam screening

A separate first-cut beam method may be used only when the route can be reduced to one qualified gravity-bending plane. This is a screening model, not a three-dimensional piping flexibility model.

| Condition     | Requirement                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| Geometry      | One ordered beam path with no branches or loops; support stations strictly increasing.                                             |
| Bending plane | One declared gravity-bending plane; out-of-plane effects excluded.                                                                 |
| Members       | Straight beam spans or declared equivalent straight segments; no elbow flexibility or tee flexibility.                             |
| Section       | Piecewise-constant E and I are allowed when each span property is explicit.                                                        |
| Supports      | Bilateral vertical translation constraints only; rotations free unless an explicit fixed-end screening boundary is declared.       |
| Loads         | Gravity distributed loads and explicit point forces only.                                                                          |
| Excluded      | Thermal strain, pressure thrust, prescribed movement, springs, gaps, friction, support settlement, lateral/axial load and torsion. |

### 5.3 Support eligibility

`verticalState = support.restraintCapability.directions.VERTICAL.state`

RESTRAINED:  
eligible within the selected screening method

FREE:  
not a vertical support

GAP:  
blocked - contact state unresolved

SPRING:  
blocked - stiffness/preload solution required

UNKNOWN or CONFLICT:  
blocked - evidence unresolved

## 6. Mass and COG calculations

### 6.1 Unit discipline

- Canonical calculation units are metres, kilograms, newtons and pascals.
- Source millimetres are converted once at the source-adaptor boundary.
- Mass remains kg or kg/m; force remains N or N/m.
- Engineering calculations are not rounded before equilibrium, COG or stress evaluation.

### 6.2 Pipe metal mass per length

Do = outer diameter [m]  
 t = analysis wall thickness for mass [m]  
 Di = Do - 2t

$A_{\text{metal}} = \pi * t * (Do - t)$   
 $m_{\text{pipe}}' = A_{\text{metal}} * \rho_{\text{material}}$

Validity:  
 Do > 0  
 t > 0  
 Di > 0  
 rho\_material > 0

The stable annulus expression  $\pi * t * (Do - t)$  is preferred to direct subtraction of fourth or squared terms where an existing governed section compiler already supplies the geometry.

### 6.3 Fluid mass per length

$A_{\text{fluid}} = \pi / 4 * Di^2$   
 $m_{\text{fluid}}' = A_{\text{fluid}} * \rho_{\text{fluid}}$

### 6.4 Insulation mass per length

Dins = Do + 2 \* tins  
 $A_{\text{insulation}} = \pi / 4 * (Dins^2 - Do^2)$   
 $m_{\text{insulation}}' = A_{\text{insulation}} * \rho_{\text{insulation}}$

### 6.5 Total distributed mass and weight

$m_{\text{total}}' =$   
 $m_{\text{pipe}}'$   
 $+ m_{\text{fluid}}'$   
 $+ m_{\text{insulation}}'$   
  
 $w_{\text{total}} = m_{\text{total}}' * g$

### 6.6 Component point mass

- A component point mass requires explicit mass and an explicit COG or qualified centre.
- A component cannot simultaneously contribute both distributed and lumped mass unless a governed split is declared.
- Valve actuators, gearboxes and instruments require their own mass and COG evidence when significant.
- Unknown component mass remains blocked; it is not inferred from pressure class alone.

### 6.7 System or path COG

$M_{\text{total}} =$   
 $\text{sum}(m_j' * L_j)$   
 $+ \text{sum}(M_k)$   
  
 $R_{\text{COG}} =$   
 $[$   
 $\text{sum}(m_j' * L_j * R_{\text{mid},j})$   
 $+ \text{sum}(M_k * R_k)$   
 $]$   
 $/ M_{\text{total}}$

COG is retained for audit, visual review, moment closure and two-support static checks. It shall not be used to solve a multi-support indeterminate structure.

## 7. First-cut gravity support-load methods

### 7.1 Method A — W10.5 simple-span tributary screening

Adjacent qualified vertical supports define independent simple screening spans. Distributed loads crossing support stations are split at those stations before reaction allocation.

#### 7.1.1 Point force

For span length  $L$  and point force  $P$  at distance  $x$  from the left support:

$$R_{\text{left}} = P * (L - x) / L$$

$$R_{\text{right}} = P * x / L$$

#### 7.1.2 Uniform distributed segment

For constant load  $w$  acting from station  $a$  to  $b$  within a support span:

$$W = w * (b - a)$$

$$x_{\text{centroid}} = (a + b) / 2$$

$$R_{\text{left}} = W * (L - x_{\text{centroid}}) / L$$

$$R_{\text{right}} = W * x_{\text{centroid}} / L$$

#### 7.1.3 Screening equilibrium

$$\text{forceResidual} =$$

$$\text{sum}(\text{screenedSupportForces})$$

$$- \text{sum}(\text{appliedGravityForces})$$

Qualification requires:

$$\text{abs}(\text{forceResidual}) \leq$$

$$\text{max}(\text{absoluteTolerance},$$

$$\text{relativeTolerance} * \text{forceScale})$$

### 7.2 Method B — qualified one-dimensional continuous-beam screening

Where a simple-chain route has three or more bilateral vertical supports and satisfies the applicability requirements in Section 5.2, a one-dimensional gravity beam screening may be used instead of empirical tributary coefficients.

**Fixed 0.4/0.6 support coefficients are prohibited. The beam method must solve joint rotations and member end moments from classical compatibility and equilibrium.**

#### 7.2.1 Slope-deflection formulation

For member  $i$ - $j$  with length  $L$ , flexural rigidity  $EI$ , joint rotations  $\theta_i$  and  $\theta_j$ , and no support settlement in this screening release:

$$M_{ij} = (2EI/L) * (2\theta_i + \theta_j) + FEM_{ij}$$

$$M_{ji} = (2EI/L) * (2\theta_j + \theta_i) + FEM_{ji}$$

At every unrestrained rotational joint:

$$\text{sum}(\text{member end moments at the joint}) = 0$$

At a declared fixed screening boundary:

$$\theta = 0$$

#### 7.2.2 Fixed-end moments

Fixed-end moments shall be generated from exact load-shape formulas or exact integration for each supported load primitive. The implementation shall retain the load primitive, formula ID, substitutions and resulting member-end terms.

#### 7.2.3 Reactions and span fields

- After joint rotations and end moments are solved, vertical reactions are recovered by member free-body equilibrium.
- Shear and bending-moment fields are evaluated along each span from the solved end actions and applied loads.
- Global vertical force and moment equilibrium are checked about a retained reference station.
- A negative vertical result is retained as a signed mathematical result. It is not interpreted as a physical hold-down or lift-off state.

#### 7.2.4 Signed-result interpretation

positive screened vertical force:  
upward support force on the screened beam

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negative screened vertical force:  
mathematical tension demand under a bilateral beam model  
-> CONDITIONAL  
-> unilateral support/contact interpretation prohibited  
-> RUN LFEA or nonlinear contact analysis

### 7.3 Method selection

| Condition                                                   | Required method                   |
|-------------------------------------------------------------|-----------------------------------|
| Two adjacent supports; simple chain                         | W10.5 simple-span screening       |
| Three or more bilateral supports; qualified 1D plane        | Continuous-beam gravity screening |
| Branch, tee, loop or spatial frame                          | RUN LFEA                          |
| Spring, gap, lift-off, friction or settlement               | RUN LFEA / nonlinear analysis     |
| Thermal, guide, line-stop, anchor or nozzle result required | RUN LFEA                          |

## 8. Sag and span screening

### 8.1 Simple-span uniform load

$\delta_{\max} = 5 w L^4 / (384 E I)$   
 $M_{\max} = w L^2 / 8$

### 8.2 Cantilever uniform load

$\delta_{\text{tip}} = w L^4 / (8 E I)$   
 $M_{\text{fixed}} = w L^2 / 2$

### 8.3 Continuous beam

For a continuous-beam screening result, deflection shall be recovered from the solved span rotation and bending field. No universal multiplier such as `1/384` or `2.1/384` may be applied to all interior or end spans.

### 8.4 Sag acceptance criterion

- The sag limit is a project screening criterion, not a universal ASME B31.3 compliance limit.
- The criterion must identify its source, revision, service basis and applicable span type.
- Exceeding the criterion produces ESCALATE, not automatic failure of the piping code.
- Drainage, slope, liquid pocketing and operability remain project engineering considerations.

## 9. Sustained stress screening indicator

**The first-cut stress result is an indicator only. It is based on simplified gravity/pressure actions and cannot represent full 3D piping flexibility, fitting flexibility, thermal expansion, occasional loading or code-qualified B31.3 combinations.**

### 9.1 Required inputs

| Input                 | Requirement                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Section               | Outer diameter, explicit evaluation thickness, area, section modulus and polar section modulus.                       |
| Actions               | First-cut axial force if available, one-dimensional gravity bending moment, and torsion only when explicitly derived. |
| Pressure contribution | User/profile supplied formula and pressure basis; not invented by the UI.                                             |
| Factors               | Caller-supplied stress indices/SIFs with source and applicability.                                                    |
| Allowable             | Caller-supplied screening allowable with material, temperature, source and revision.                                  |

### 9.2 Generic screening arithmetic

$S_{\text{axial}} = (N / A) * I_{\text{axial}}$   
 $S_{\text{bending}} =$

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```
sqrt(  
  (i_in_plane * M_in_plane / Z)^2  
  +  
  (i_out_plane * M_out_plane / Z)^2  
)  
  
S_torsion =  
  i_torsion * T / Zp  
  
S_screen =  
  abs(S_pressure + S_axial)  
  +  
  sqrt(S_bending^2 + S_torsion^2)  
  
utilization_screen =  
  S_screen / S_allowable_screen
```

### 9.3 Restrictions

- No default corrosion allowance, mill tolerance or analysis thickness reduction.
- No embedded ASME allowable, SIF, stress-index or cycle-factor table.
- No thermal expansion stress or lift-off multiplier.
- No claim that a planar first-cut moment is the governing in-plane/out-of-plane code moment.
- No code result when factor applicability, allowable source or thickness basis is unresolved.

### 9.4 Status wording

SCREENING WITHIN DECLARED METHOD  
SCREENING EXCEEDS PROJECT LIMIT  
CONDITIONAL - LFEA REQUIRED  
BLOCKED - INPUT OR APPLICABILITY UNRESOLVED

Prohibited:  
ASME B31.3 PASS  
ASME B31.3 FAIL  
CODE COMPLIANT

## 10. Manual support-unavailable sensitivity

The first-cut method may provide a manually declared sensitivity scenario in which one or more supports are excluded from the gravity screening model. This is useful for understanding potential span sensitivity, but it is not a lift-off prediction.

- The user explicitly selects each unavailable support.
- The scenario reason is retained, for example `MANUAL\_SUPPORT\_UNAVAILABLE\_SENSITIVITY`.
- The modified topology is requalified under the same screening rules.
- Reactions, moments and sag are recomputed from the modified gravity model.
- No thermal cause, contact state or probability is inferred.
- The result label includes `USER-DECLARED SENSITIVITY`.

## 11. Pre-calculation audit and assumptions

### 11.1 Audit dimensions

| Audit area       | Blocking conditions                                                                             |
|------------------|-------------------------------------------------------------------------------------------------|
| Topology         | Disconnected path, branch, loop, unsupported overhang, duplicate station or invalid ordering.   |
| Geometry         | Missing OD, wall, section, length or invalid inner diameter.                                    |
| Mass             | Missing material density, fluid density, insulation or component mass/COG.                      |
| Supports         | Unattached, ambiguous, FREE, GAP, SPRING, UNKNOWN or CONFLICT vertical state.                   |
| Beam screening   | Missing E/I, unsupported geometry, load type or boundary assumption.                            |
| Stress screening | Missing pressure basis, factor source, allowable source, evaluation thickness or applicability. |

### 11.2 Assumption workflow

User requests first-cut calculation  
-> run completeness audit  
-> show unresolved and proposed assumptions  
-> user returns to enrichment  
    OR accepts a traceable approximation

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- > seal assumption set
- > create calculation package
- > execute selected screening method
- > present results and limitations

### 11.3 Assumption record

```
{
  "schema": "first-cut-assumption-set/v1",
  "sourceSemanticHash": "fnv1a64:...",
  "profileSemanticHash": "fnv1a64:...",
  "assumptions": [
    {
      "assumptionId": "ASM-0001",
      "entityId": "PIPE-019",
      "fieldId": "fluidDensity",
      "value": 850.0,
      "unit": "kg/m3",
      "source": "PROJECT-PROCESS-BASIS-REV2",
      "reason": "Operating fluid density not present in source model",
      "approver": "USER-DECLARED",
      "limitations": ["OPE case only"]
    }
  ],
  "semanticHash": "fnv1a64:...",
  "evidenceHash": "fnv1a64:..."
}
```

### 11.4 No source mutation

- Imported source entities are never modified by calculation.
- Assumptions and overrides reside in separate immutable packages.
- Execution time and user-interface metadata do not enter semantic identity.
- A changed source, profile, override or assumption invalidates all dependent first-cut results.

## 12. Result contracts and traceability

| Contract                             | Authority                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------|
| first-cut-load-estimation-profile/v1 | Method selection, gravity, tolerances, sag criterion, permitted load types and escalation rules. |
| first-cut-assumption-set/v1          | Accepted project assumptions and evidence.                                                       |
| first-cut-mass-ledger/v1             | Distributed and lumped masses, COG, sources and double-count checks.                             |
| first-cut-support-screening/v1       | W10.5 simple-span support shares and equilibrium.                                                |
| first-cut-beam-screening/v1          | Qualified continuous-beam reactions, end moments, span fields and deflection.                    |
| first-cut-sustained-screening/v1     | Screening stress terms, factors, allowable and limitations.                                      |
| first-cut-calculation-package/v1     | Parent hashes, selected method, all results, diagnostics and status.                             |

### 12.1 Minimum lineage

```
sourceSemanticHash
-> enrichmentResultSemanticHash
-> modelLoadPrimitiveSemanticHash
-> topology/pathSemanticHash
-> assumptionSetSemanticHash
-> screeningProfileSemanticHash
-> massLedgerSemanticHash
-> supportOrBeamScreeningSemanticHash
-> sustainedScreeningSemanticHash
-> calculationPackageSemanticHash
-> evidenceHash
```

### 12.2 Support result record

```
{
  "supportId": "SUP-042",
  "loadCaseId": "OPE",
  "resultKind": "SCREENED_GRAVITY_SHARE",
  "methodId": "SIMPLE_SPAN_TRIBUTARY_VERTICAL_V1",
  "screenedVerticalForceN": 2268.7,
  "globalForce": null,
  "globalMoment": null,
  "thermalLoad": null,
}
```



```

"guideLoad": null,
"lineStopLoad": null,
"anchorLoad": null,
"nozzleLoad": null,
"status": "SCREENING_WITHIN_DECLARED_METHOD",
"limitations": [
  "NO_THERMAL_REACTION",
  "NO_LATERAL_OR_AXIAL_REACTION",
  "NOT_B31_3_COMPLIANCE"
]
}

```

## 13. Calculation-basis panel

### 13.1 Required sections

| Panel section     | Content                                                                         |
|-------------------|---------------------------------------------------------------------------------|
| Header            | Support ID, source line/system, method, case, status and source hashes.         |
| Applicability     | Topology and support qualification; blockers and assumptions.                   |
| Contributing mass | Pipe, fluid, insulation and component ledger with source and affected span.     |
| Load allocation   | Span endpoints, support stations, point-load shares and distributed segments.   |
| Beam result       | When applicable: end moments, reaction, span moment and deflection.             |
| Sag screening     | Calculated sag, project criterion, utilization and source.                      |
| Stress indicator  | Terms, factors, allowable, utilization and non-compliance warning.              |
| Not evaluated     | Thermal, guide, line-stop, anchor, nozzle, spring/contact and expansion stress. |
| Evidence          | Parent hashes, profile, assumption set and export actions.                      |

### 13.2 Example callout

#### FIRST-CUT SCREENING

Support: SUP-042

Case: OPE

Method: W10.5 simple span

Screened vertical share: 2.27 kN

Sag screening: 1.42 mm / 2.50 mm project criterion

Sustained indicator: 0.47

Thermal: NOT EVALUATED - RUN LFEA

Guide: NOT EVALUATED - RUN LFEA

Line stop: NOT EVALUATED - RUN LFEA

Anchor moments: NOT EVALUATED - RUN LFEA

Nozzle loads: NOT EVALUATED - RUN LFEA

## 14. Implementation modules

| Module                                                      | Responsibility                                            |
|-------------------------------------------------------------|-----------------------------------------------------------|
| src/core/first-cut-load-estimation/contracts.js             | Schemas, keys, statuses and exact validation.             |
| src/core/first-cut-load-estimation/profile.js               | Method profile, tolerances, sag criterion and escalation. |
| src/core/first-cut-load-estimation/mass-ledger.js           | Mass, weight and COG compilation.                         |
| src/core/first-cut-load-estimation/beam-screening.js        | Optional slope-deflection gravity beam screening.         |
| src/core/first-cut-load-estimation/sag-recovery.js          | Span deflection and governing sag.                        |
| src/core/first-cut-load-estimation/sustained-screening.js   | Non-code screening indicator.                             |
| src/core/first-cut-load-estimation/package.js               | Parent binding and final calculation package.             |
| src/workspace/enrichment/precac-audit-controller.js         | Audit and assumption workflow.                            |
| src/workspace/enrichment/support-calculation-basis-panel.js | Read-only result and evidence presentation.               |
| src/workspace/sequential-sketcher/support-load-presenter.js | Consumes sealed screening/LFEA results; no mechanics.     |

### 14.1 Module boundary

- Core calculation modules shall not import workspace, SVG, WebGL or DOM code.
- Workspace modules shall not calculate mass, beam reactions, sag or stress.

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- W10.4 and W10.5 outputs shall be consumed rather than recomputed.
- The support presenter shall prefer a qualified LFEA reaction when available and otherwise show the first-cut screening with its exact label.

## 15. Test and benchmark requirements

### 15.1 Contract and evidence tests

- FC-T01 exact schemas accepted
- FC-T02 extra keys rejected
- FC-T03 deep immutability
- FC-T04 stale parent hash rejected
- FC-T05 ABSENT distinct from PRESENT\_NULL
- FC-T06 direct zero retained
- FC-T07 caller arrays are not mutated
- FC-T08 input order does not change semantic identity
- FC-T09 Unicode evidence hashes deterministically
- FC-T10 timestamps excluded from semantic identity

### 15.2 Mass and COG benchmarks

| ID       | Benchmark                         | Acceptance                                |
|----------|-----------------------------------|-------------------------------------------|
| MASS-01  | Pipe annulus mass                 | Closed-form relative error $\leq 1e-12$ . |
| MASS-02  | Fluid mass                        | Closed-form relative error $\leq 1e-12$ . |
| MASS-03  | Insulation annulus                | Closed-form relative error $\leq 1e-12$ . |
| MASS-04  | Point component mass              | Exact source value retained.              |
| MASS-05  | Mixed distributed/lumped COG      | Mass and first-moment closure.            |
| MASS-06  | Explicit zero fluid               | Accepted and distinct from missing.       |
| MASS-B01 | Invalid inner diameter            | Blocked.                                  |
| MASS-B02 | Component mass without COG/centre | Blocked.                                  |
| MASS-B03 | Distributed/lumped double count   | Blocked.                                  |

### 15.3 Simple-span benchmarks

| ID            | Benchmark                 | Acceptance                                                 |
|---------------|---------------------------|------------------------------------------------------------|
| SPAN-UDL-01   | Uniform load              | $R_{\text{left}} = R_{\text{right}} = wL/2$ .              |
| SPAN-POINT-01 | Point force               | $R_{\text{left}} = P(L-x)/L$ ; $R_{\text{right}} = Px/L$ . |
| SPAN-MIXED-01 | UDL plus point force      | Superposition and force equilibrium.                       |
| SPAN-SPLIT-01 | Load crossing support     | Exact split at support station.                            |
| SPAN-B01      | Overhang outside envelope | Blocked.                                                   |
| SPAN-B02      | Tee or loop topology      | Blocked.                                                   |
| SPAN-B03      | Spring or gap support     | Blocked.                                                   |

### 15.4 Continuous-beam benchmarks

| ID       | Benchmark                            | Acceptance                                                                            |
|----------|--------------------------------------|---------------------------------------------------------------------------------------|
| BEAM-01  | Single simple span UDL               | Reactions $wL/2$ ; $M_{\text{max}} wL^2/8$ ; sag $5wL^4/(384EI)$ .                    |
| BEAM-02  | Cantilever UDL                       | Reaction $wL$ ; fixed moment $wL^2/2$ ; tip sag $wL^4/(8EI)$ .                        |
| BEAM-03  | Two equal continuous spans, both UDL | Outer reactions $3wL/8$ each; interior reaction $5wL/4$ ; interior moment $-wL^2/8$ . |
| BEAM-04  | Unequal spans and EI                 | Independent slope-deflection reference.                                               |
| BEAM-05  | Point load in one span               | Independent hand/matrix solution.                                                     |
| BEAM-06  | Global force and moment closure      | Within profile tolerances.                                                            |
| BEAM-B01 | Negative bilateral reaction          | Conditional; no lift-off conclusion.                                                  |
| BEAM-B02 | Branch or out-of-plane route         | Blocked and escalated.                                                                |

## 15.5 Sag and stress screening tests

| ID      | Benchmark                             | Acceptance                                     |
|---------|---------------------------------------|------------------------------------------------|
| SAG-01  | Simple span                           | Closed-form deflection.                        |
| SAG-02  | Continuous beam                       | Recovered field against independent reference. |
| SAG-B01 | Missing project criterion             | Sag calculated; acceptance status unresolved.  |
| STR-01  | Fictional sustained screening example | Independent hand calculation.                  |
| STR-02  | Factor direction separation           | No scalar factor collapse.                     |
| STR-B01 | Missing allowable source              | Blocked.                                       |
| STR-B02 | Thermal input supplied                | Rejected by first-cut screening package.       |
| STR-B03 | Code-compliance status requested      | Rejected.                                      |

## 15.6 UI end-to-end tests

- UI-FC-01 open, dock, collapse, close and restore
- UI-FC-02 staged assumptions survive panel close
- UI-FC-03 blocked audit highlights exact entities
- UI-FC-04 apply assumptions creates a separate package
- UI-FC-05 source bytes remain unchanged
- UI-FC-06 method label is always visible
- UI-FC-07 screening is never labelled reaction
- UI-FC-08 NOT EVALUATED fields remain null
- UI-FC-09 stale result cannot display as current
- UI-FC-10 evidence action shows parent hashes
- UI-FC-11 keyboard and screen-reader access
- UI-FC-12 export/re-import is deterministic

## 16. Anti-drift controls

- Reject hardcoded OD, wall, span, fluid density, insulation density, temperature, allowable or SIF values.
- Reject fixed 0.4/0.6 continuous-beam coefficients.
- Reject the exact 4x lift-off multiplier.
- Reject guide, line-stop, anchor or nozzle load fields with numerical values in a first-cut package.
- Reject thermal strain, prescribed movement, spring, gap or friction imports into the first-cut solver.
- Reject the words 'B31.3 PASS', 'B31.3 FAIL' or 'CODE COMPLIANT' in first-cut result presentation.
- Reject source-entity mutation by the assumption or calculation workflow.
- Reject workspace-side mass, beam, sag or stress calculations.
- Reject locale-dependent ordering, random IDs and timestamps in semantic hashes.
- Reject stale result export.

```
const forbiddenRules = [
  {
    code: 'FIRST_CUT_THERMAL_MECHANICS_PROHIBITED',
    root: 'src/core/first-cut-load-estimation',
    pattern: /thermalReaction|liftOff|guideLoad|lineStopLoad|nozzleLoad/u,
  },
  {
    code: 'EMPIRICAL_CONTINUOUS_COEFFICIENT_PROHIBITED',
    root: 'src',
    pattern: /0\.4\s*\*|0\.6\s*\*|four\s*\times\s*\lift/i,
  },
  {
    code: 'FIRST_CUT_CODE_COMPLIANCE_CLAIM',
    root: 'src/workspace',
    pattern: /B31\.3\s*(PASS|FAIL)|CODE\s+COMPLIANT/iu,
  },
  {
    code: 'WORKSPACE_MECHANICS_PROHIBITED',
    root: 'src/workspace',
    pattern: /slopeDeflection|sectionModulus.*moment|calculateSag|calculateStress/u,
  }
];
```

## 17. Escalation to LFEA

| Trigger | Required escalation |
|---------|---------------------|
|---------|---------------------|

## First-Cut Piping Load Estimation | Screening Without LFEA

| Trigger                                                | Required escalation                                                 |
|--------------------------------------------------------|---------------------------------------------------------------------|
| Thermal expansion, contraction or imposed movement     | LFEA physical thermal/prescribed-movement load cases.               |
| Guide or line-stop load                                | LFEA solved reactions resolved in support-local directions.         |
| Anchor force and moment                                | LFEA six-DOF reaction recovery.                                     |
| Nozzle force and moment                                | LFEA equipment-interface recovery in nozzle-local axes.             |
| Branch, loop or spatial 3D routing                     | LFEA full 3D frame model.                                           |
| Bend, tee or reducer flexibility important             | LFEA B-3.2 piping-component mechanics.                              |
| Gap, lift-off, one-way rest or friction                | Future nonlinear LFEA contact package.                              |
| Linear spring or foundation stiffness                  | LFEA linear spring/constraint solve.                                |
| Support settlement                                     | LFEA prescribed movement.                                           |
| B31.3 sustained, occasional or displacement acceptance | LFEA B-3.4 recovery and B-4.0 code engine.                          |
| Equipment allowable comparison                         | LFEA reaction/nozzle-load export plus authorized allowable profile. |

## 18. Release gates

```

npm ci
npm run check:first-cut-load-estimation
npm run check:first-cut-beam-screening
npm run check:first-cut-sustained-screening
npm run check:first-cut-anti-drift
npm run check:shared-analysis
npm run check:workspace-contracts
npm run check:lfea-svg
npm run check:sequential-sketcher
npm run check:lfea-workbench
npm run check:e2e
npm run syntax:strict
npm run check:imports
npm run build
npm run gate
git diff --check
test -z "$(git status --short)"

```

## 19. Final acceptance criteria

1. Every input is explicit, authorized or blocked.
2. Mass and COG close against independent first moments.
3. W10.5 simple-span results remain within the existing topology and support scope.
4. The optional beam method uses compatibility/equilibrium and no empirical support coefficients.
5. Every gravity result passes force equilibrium; beam results also pass moment equilibrium.
6. Sag uses the correct method-specific field and a project-supplied criterion.
7. The stress result is explicitly a screening indicator and retains every factor and source.
8. Thermal, guide, line-stop, anchor, nozzle, spring/contact and code-compliance results remain unevaluated.
9. The UI presents method, case, assumptions, limitations and parent identities.
10. A changed source, override, assumption or profile invalidates the result.
11. All deliberate regressions and exact-head release gates pass.

## 20. Final authority statement

**FIRST-CUT AUTHORITY:** mass and COG; EMPTY/OPE/HYD gravity primitives; W10.5 simple-span screened vertical shares; qualified one-dimensional gravity beam screening; sag screening; and a non-code sustained-stress screening indicator. All thermal, guide, line-stop, anchor, nozzle, gap/contact, spring, occasional, expansion and code-compliance results require LFEA.

## Appendix A — Calculation status matrix

| Status              | Meaning                                                                                              |
|---------------------|------------------------------------------------------------------------------------------------------|
| QUALIFIED SCREENING | Method and data are within the declared first-cut profile.                                           |
| CONDITIONAL         | Result is mathematically available but a physical interpretation or applicability condition requires |

## First-Cut Piping Load Estimation | Screening Without LFEA

| Status   | Meaning                                                            |
|----------|--------------------------------------------------------------------|
|          | LFEA.                                                              |
| ESCALATE | The requested quantity lies outside first-cut authority.           |
| BLOCKED  | Required evidence, topology or method applicability is unresolved. |
| STALE    | A parent source, override, assumption or profile changed.          |

## Appendix B — Revised support basis panel fields

| Field                 | Required display                                                                            |
|-----------------------|---------------------------------------------------------------------------------------------|
| Method                | W10.5 simple span or first-cut continuous beam.                                             |
| Case                  | EMPTY, OPE or HYD.                                                                          |
| Source evidence       | Geometry, section, material, contents, insulation, component masses and support capability. |
| Mass ledger           | Each contributing distributed or point mass.                                                |
| Support/path geometry | Stations, span lengths and path qualification.                                              |
| Calculated values     | Screened vertical share or beam vertical force, moment and sag.                             |
| Stress indicator      | Optional pressure/gravity screening terms and allowable source.                             |
| Limitations           | Explicit NOT EVALUATED quantities.                                                          |
| Escalation            | Direct action to prepare/run LFEA.                                                          |

## Appendix C — Example calculation package

```
{
  "schema": "first-cut-calculation-package/v1",
  "sourceSemanticHash": "fnv1a64:...",
  "enrichmentResultSemanticHash": "fnv1a64:...",
  "loadPrimitiveSemanticHash": "fnv1a64:...",
  "pathSemanticHash": "fnv1a64:...",
  "assumptionSetSemanticHash": "fnv1a64:...",
  "profileSemanticHash": "fnv1a64:...",
  "method": "SIMPLE_SPAN_TRIBUTARY_VERTICAL_V1",
  "loadCaseId": "OPE",
  "massLedger": { "...": "..."},
  "supportScreening": { "...": "..."},
  "beamScreening": null,
  "sustainedScreening": { "...": "..."},
  "notEvaluated": [
    "THERMAL_SUPPORT_LOAD",
    "GUIDE_REACTION",
    "LINE_STOP_REACTION",
    "ANCHOR_SIX_DOF_LOAD",
    "NOZZLE_LOAD",
    "B31_3_COMPLIANCE"
  ],
  "status": "QUALIFIED_SCREENING",
  "semanticHash": "fnv1a64:...",
  "evidenceHash": "fnv1a64:..."
}
```

## Appendix D — Source basis

This note consolidates and corrects the user-supplied Technical Audit and revised Appendices B-D for Bulk Data Enrichment, tributary support-load estimation, continuous-beam sag, calculation-basis presentation and pre-calculation auditing. The thermal/contact and full interface-load portions have been separated into the independent LFEA concept note.